

Technical Challenge Data Analysis

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July 2, 2026

Recruitment Funnel Analysis – Executive Summary

1. Approach

The objective of this analysis was to identify the factors that most influence candidate progression throughout the recruitment funnel and to provide actionable recommendations for improving hiring efficiency.

The analysis followed a structured, data-driven approach:

- Performed an exploratory analysis of the dataset (601 candidates and 35 variables) to understand data quality and key attributes.
- Examined the recruitment funnel from sourcing to hiring.
- Calculated conversion rates for each stage of the hiring process.
- Compared sourcing channels based on hiring performance.
- Identified behavioral and profile characteristics associated with higher hiring probability.
- Investigated patterns related to recruiters, candidate engagement, seniority, and assessment performance.
- Translated quantitative findings into practical recommendations for the Talent Acquisition team.

Rather than focusing only on descriptive statistics, the analysis aimed to uncover business insights that could support better recruiting decisions.

Decisions Made

Several analytical decisions were taken to ensure consistency and business relevance.

Conversion as the primary KPI

The main performance metric used throughout the analysis was the end-to-end hire conversion rate, allowing direct comparison across sourcing channels and candidate profiles.

Focus on funnel progression

Instead of evaluating only final hiring outcomes, each recruitment stage was analyzed separately to identify where candidates were most likely to advance or drop off.

Business-oriented interpretation

Whenever possible, findings were translated into operational recommendations instead of simply reporting numerical results.

Use of AI

Artificial Intelligence was used as a productivity and analytical support tool throughout this project with an specific prompt.

AI assisted with:

- Structuring the exploratory data analysis.
- Identifying patterns and correlations within the recruitment funnel.
- Summarizing quantitative findings into business insights.
- Supporting hypothesis generation for possible causes behind observed trends.
- Improving the clarity, organization, and professional writing of the final report.

All conclusions and recommendations were validated against the dataset, ensuring that insights remained evidence-based rather than generated solely by AI.

AI was therefore used to accelerate analysis and enhance communication, while analytical judgment and interpretation remained under my responsibility.

Key Insights

❑ **High-performing sourcing channels**

GitHub, Inbound, and Hunting consistently delivered the highest hire conversion rates, indicating that these channels produce higher-quality candidates.

❑ **Candidate engagement is a leading indicator**

Candidates who responded to recruiter outreach were significantly more likely to progress through the hiring funnel.

❑ **Assessment performance predicts hiring success**

Candidates who ultimately received offers consistently achieved higher technical, behavioral, and hiring manager evaluation scores.

❑ **Seniority positively impacts conversion**

More experienced candidates showed higher hiring rates, although successful hires occurred across all experience levels.

Key Insights

Funnel optimization offers the greatest opportunity

The largest opportunities for improvement lie in:

- Increasing response rates through personalized outreach.
 - Improving early candidate qualification.
 - Investing more heavily in high-performing sourcing channels.
 - Continuously monitoring recruitment KPIs to enable data-driven decision making.
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Final Considerations

This analysis demonstrates how recruitment data can support more strategic Talent Acquisition decisions. Rather than measuring success solely by hiring volume, tracking funnel conversion, candidate engagement, sourcing effectiveness, and assessment performance enables recruiters to prioritize high-potential candidates, allocate resources more efficiently, and continuously optimize hiring outcomes.

Adopting a data-driven recruitment approach allows Talent Acquisition teams to improve both operational efficiency and the overall quality of hires while creating a more consistent and scalable hiring process.

Future Opportunities

Given additional historical hiring data, the next step would be to develop predictive recruitment dashboards and machine learning models capable of estimating each candidate's probability of advancing through the hiring funnel. This would enable recruiters to prioritize high-potential candidates, optimize sourcing investments, and make faster, more data-informed hiring decisions.

Thank you!

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